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i) The force due to surface tension acting towards right around the rim of length $2\pi R$ is $F_T = 2\pi RT$

ii) The force due to outside pressure P_1 is to the right acting across a cross sectional area of πR^2 is $F_{P_1} = P_1\pi R^2$

iii) The force due to pressure P_2 inside the bubble, acting to the left is $F_{P_2} = P_2\pi R^2$.

As the air bubble is in equilibrium under the action of these forces, $F_{P_2} = F_T + F_{P_1}$

$$P_2\pi R^2 = 2\pi RT + P_1\pi R^2$$

$$\Rightarrow (P_2 - P_1)\pi R^2 = 2\pi RT$$

Excess pressure is $\Delta P = P_2 - P_1 = \frac{2T}{R}$ (7.30)

2) Excess pressure inside a soap bubble

Consider a soap bubble of radius R and the surface tension of the soap bubble be T as shown in Figure 7.28 (b). A soap bubble has two liquid surfaces in contact with air, one inside the bubble and other outside the bubble. Therefore, the force on the soap bubble due to surface tension is $2\times2\pi RT$. The various forces acting on the soap bubble are,

i) Force due to surface tension $F_T = 4\pi RT$ towards right

ii) Force due to outside pressure, $F_{P_1} = P_1\pi R^2$ towards right

iii) Force due to inside pressure, $F_{P_2} = P_2\pi R^2$ towards left

As the bubble is in equilibrium, $F_{P_2} = F_T + F_{P_1}$

$$P_2\pi R^2 = 4\pi RT + P_1\pi R^2$$

$$\Rightarrow (P_2 - P_1)\pi R^2 = 4\pi RT$$

Excess pressure is $\Delta P = P_2 - P_1 = \frac{4T}{R}$ (7.31)

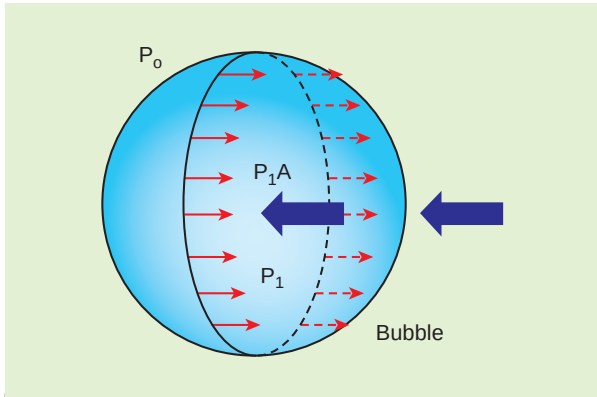


Figure7.28 (b) Soap bubble

3) Excess pressure inside the liquid drop

Consider a liquid drop of radius R and the surface tension of the liquid is T as shown in Figure 7.28 (c).

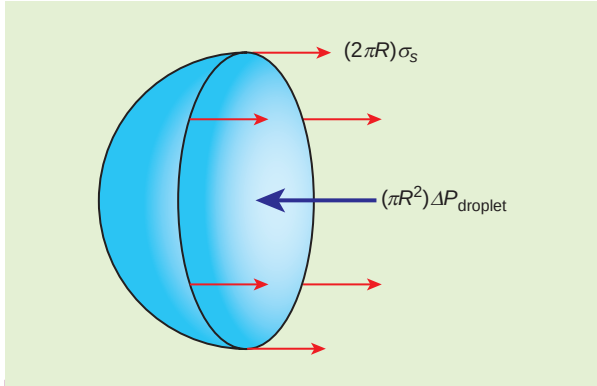


Figure7.28 (c) Liquid drop

The various forces acting on the liquid drop are,

i) Force due to surface tension $F_T = 2\pi RT$ towards right

ii) Force due to outside pressure, $F_{P_1} = P_1\pi R^2$ towards right

iii) Force due to inside pressure, $F_{P_2} = P_2\pi R^2$ towards left

As the drop is in equilibrium, $F_{P_2} = F_T + F_{P_1}$

$$P_2\pi R^2 = 2\pi RT + P_1\pi R^2$$

$$\Rightarrow (P_2 - P_1)\pi R^2 = 2\pi RT$$

Excess pressure is $\Delta P = P_2 - P_1 = \frac{2T}{R}$ (7.32)